

BEE 208 — Question 17: Stop Insulation Test Using Break

Control Statement: for loop with break | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables resistance (double) and i (int)

Step 3: Use a for loop from i = 1 to 10

Step 4: Prompt user to enter insulation resistance reading i

Step 5: Read resistance

Step 6: If resistance < 1, display 'Insulation failure detected. Test stopped.' and break

Step 7: Else display 'Insulation reading acceptable'

Step 8: End loop

Step 9: End

2. PSEUDOCODE

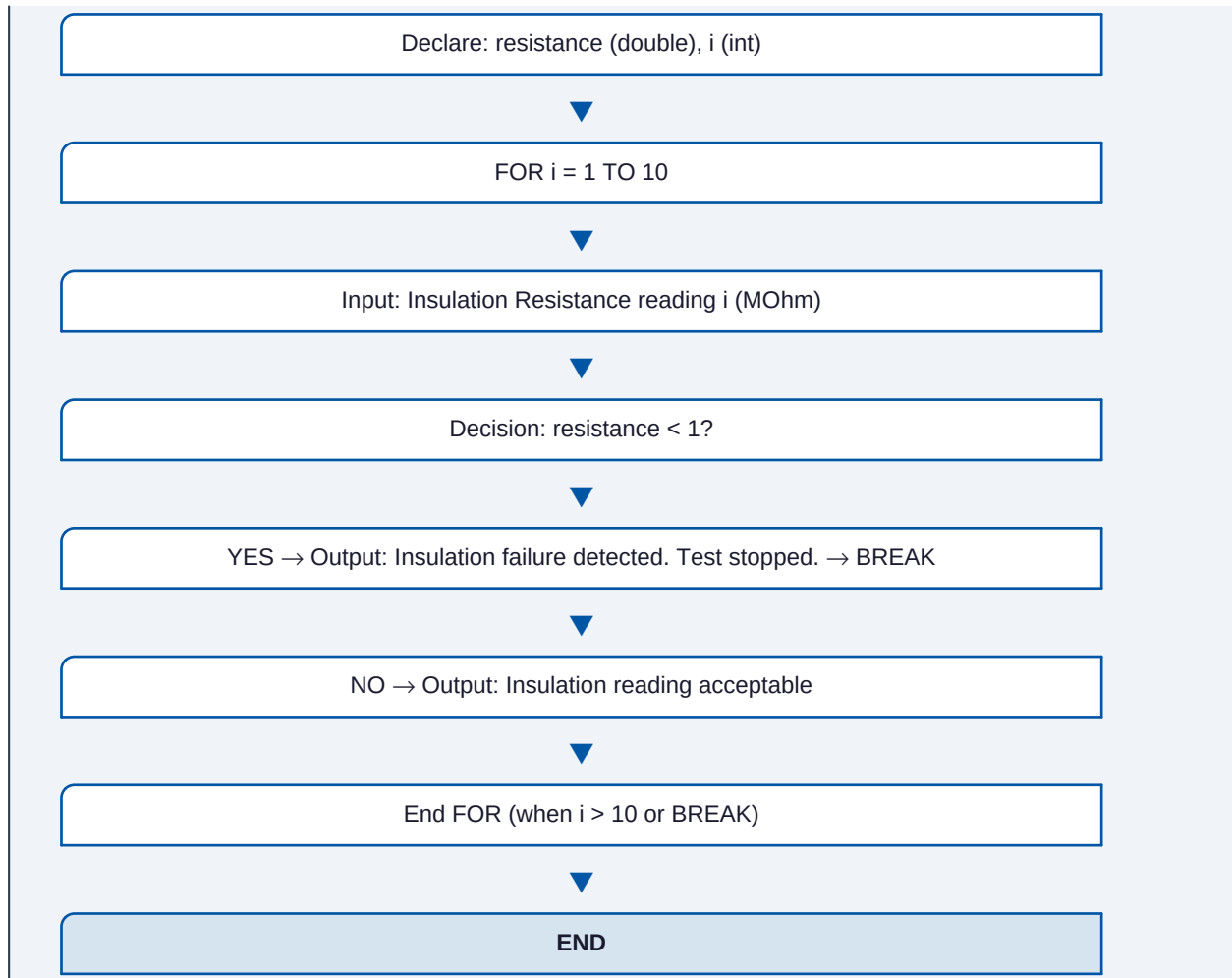
```
BEGIN  
  
DECLARE resistance AS DOUBLE  
  
DECLARE i AS INTEGER  
  
FOR i = 1 TO 10 DO  
  
    OUTPUT 'Reading', i, '- Insulation Resistance (MOhm):'  
  
    INPUT resistance  
  
    IF resistance < 1 THEN  
  
        OUTPUT 'Insulation failure detected. Test stopped.'  
  
        BREAK  
  
    ELSE  
  
        OUTPUT 'Insulation reading acceptable:', resistance, 'MOhm'  
  
    END IF  
  
END FOR  
  
END
```

3. FLOWCHART (Structured Diagram)

START



A light blue rounded rectangular box containing the word "START" in black capital letters. Below the box is a small blue downward-pointing triangle.



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	Readings: 2, 3, 0.5	Expected Output	OK×2, then: Insulation failure detected. Test stopped.
Input	10 readings all >= 1	Expected Output	10 × Insulation reading acceptable